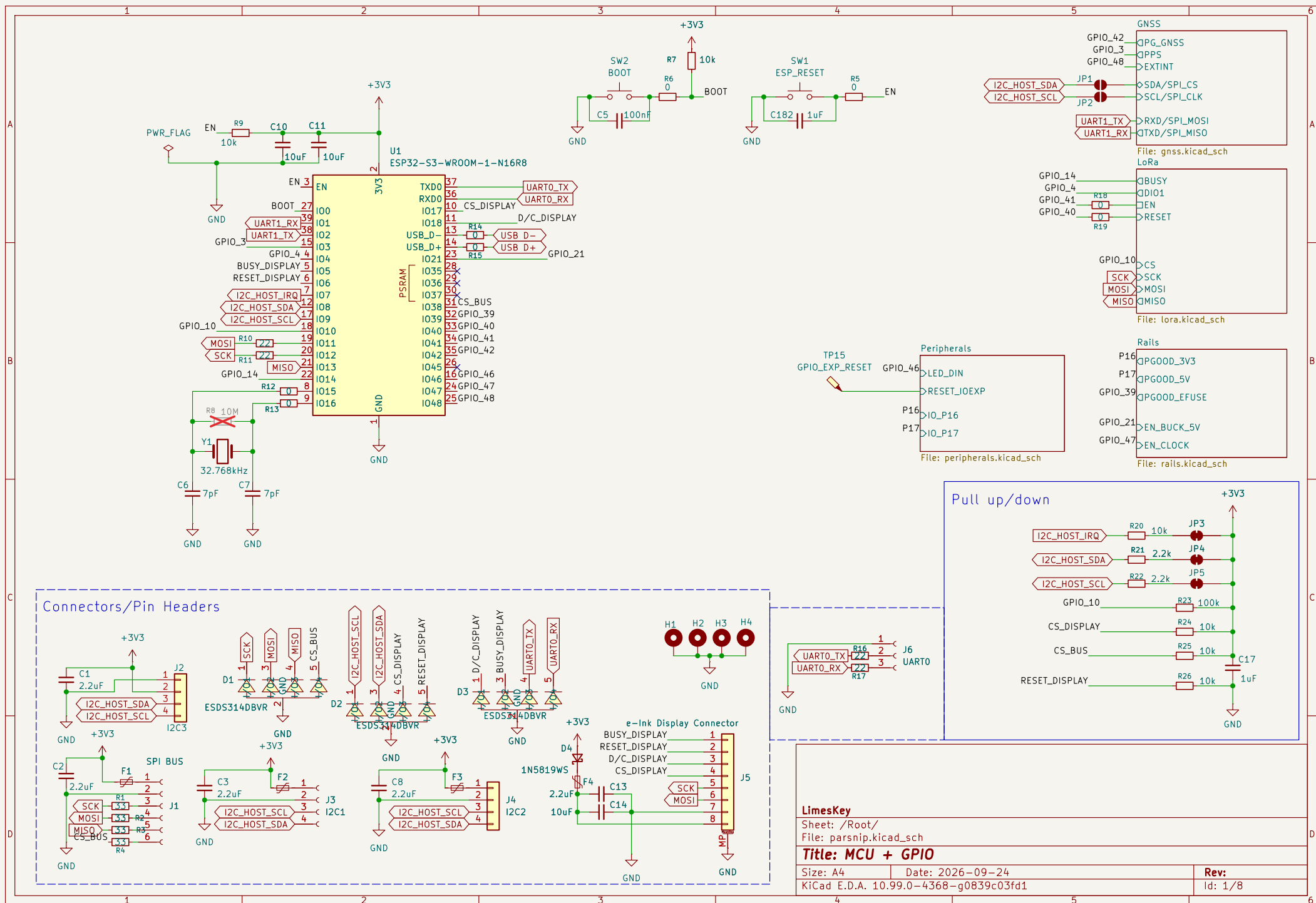




LimesKey

Sheet: /Root/

File: parsnip.kicad_sch

Title: MCU + GPIO

Size: A4

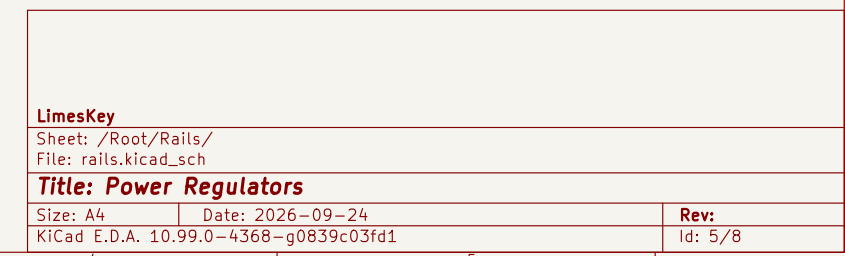
Date: 2026-09-24

KiCad E.D.A. 10.99.0-4368-g0839c03fd1

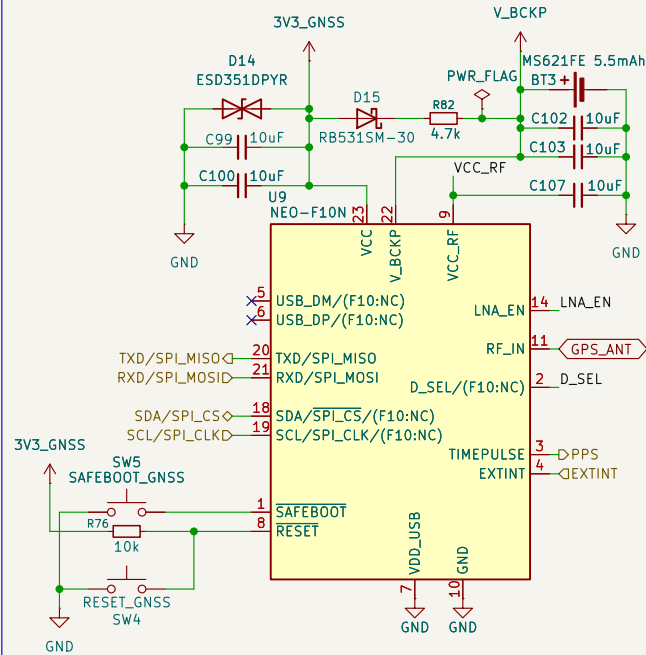
Rev:

Id: 1/8

3V3 Buck



High to enable I2C/UART, low to enable SPI
Only for NEO-M(♯)N



Sparkfun Schematic: https://cdn.sparkfun.com/assets/1/0/a/8/a/SparkFun_GPS_NEO_M9N_SMA_UPDATE.pdf
 U-Blox Datasheet: https://content.u-blox.com/sites/default/files/NEO-M9N_Integrationmanual_UBX-19014286.pdf

The schematic diagram illustrates the power management section of the TP5000 module, featuring two LDOs (U10 and U11) and various passive components.

U10: TPS7A2033PDBVR (High PSRR & Low Noise LDO)

- Input: 2.2uF capacitor (C104) connected to the IN pin.
- EN pin: Connected to the 3V3 supply.
- OUT pin: Connected to the 3V3 output, with a 10uF capacitor (C106) and a diode (D17) in parallel to ground.
- NC pin: Connected to ground.
- GND pin: Connected to ground.

U11: TPS259474LRPWR (Optional module power input protection)

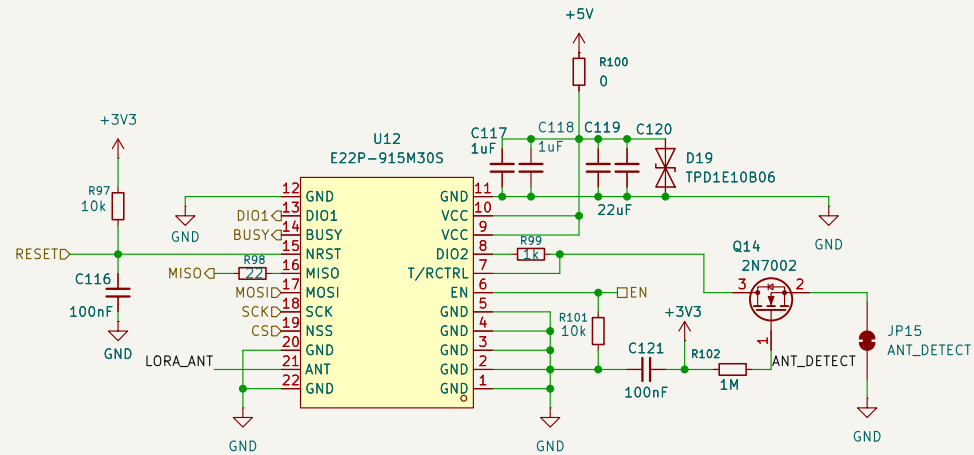
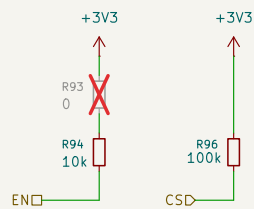
- IN pin: Connected to the 3V3 supply, with a 2.2nF capacitor (C108) to ground.
- EN/UVLO pin: Connected to the 3V3 supply.
- OVLO/OVCSEL pin: Connected to the 3V3 supply.
- PGTH/FLT pin: Connected to the 3V3 supply.
- PG/AUXOFF pin: Connected to the 3V3 supply.
- OUT pin: Connected to the 3V3 output, with a 10uF capacitor (C109) and a diode (D17) in parallel to ground.
- GND pin: Connected to ground.

Other Components:

- R84:** 150k, 0.1% resistor connected between the 3V3 supply and the 3V3 output.
- R85:** 215k, 0.1% resistor connected between the 3V3 supply and the 3V3 output.
- R87:** 113k, 0.1% resistor connected between the 3V3 supply and the 3V3 output.
- R88:** 6.65k, 1% resistor connected between the 3V3 supply and the 3V3 output.
- R89:** 200k, 0.1% resistor connected between the 3V3 supply and the 3V3 output.
- R90:** 301k resistor connected between the 3V3 supply and the 3V3 output.
- R91:** 100k resistor connected between the 3V3 supply and the 3V3 output.
- C110:** 10uF capacitor connected between the 3V3 supply and the 3V3 output.
- C111:** 10uF capacitor connected between the 3V3 supply and the 3V3 output.
- D17:** TPD1E10B06 diode connected between the 3V3 supply and the 3V3 output.
- DP_GNSS:** Signal pin connected to the 3V3 supply.
- PWR_FLAG:** Signal pin connected to the 3V3 supply.
- +3V3:** Power supply pin connected to the 3V3 supply.
- 3V3_GNSS:** Power supply pin connected to the 3V3 supply.
- DISABLE_GNSS_PROT:** Signal pin connected to the 3V3 supply.

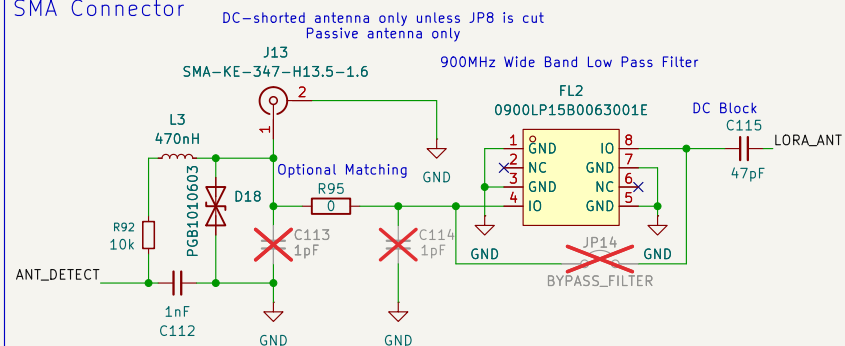
Overcurrent Threshold: 575mA

Rev: 1.1
Id: 6/8



Ensure you set DIO2 as RF Switch in program
https://jgromes.github.io/RadioLib/class_s_x126x.html#ae46e08d579f4acbad029b4cd4f4ffaf

SMA Connector



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Sheet: /Root/LoRa/
 File: lora.kicad_sch

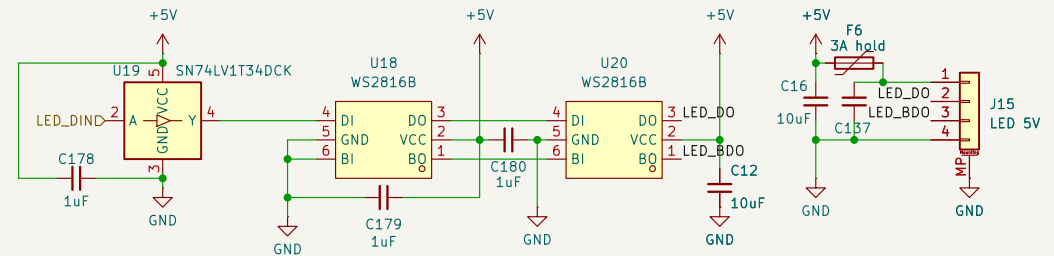
Title: LoRa Module / LoRa Antenna

Size: A4 Date: 2026-08-24
 KiCad E.D.A. 10.99.0-4368-g0839c03fd1

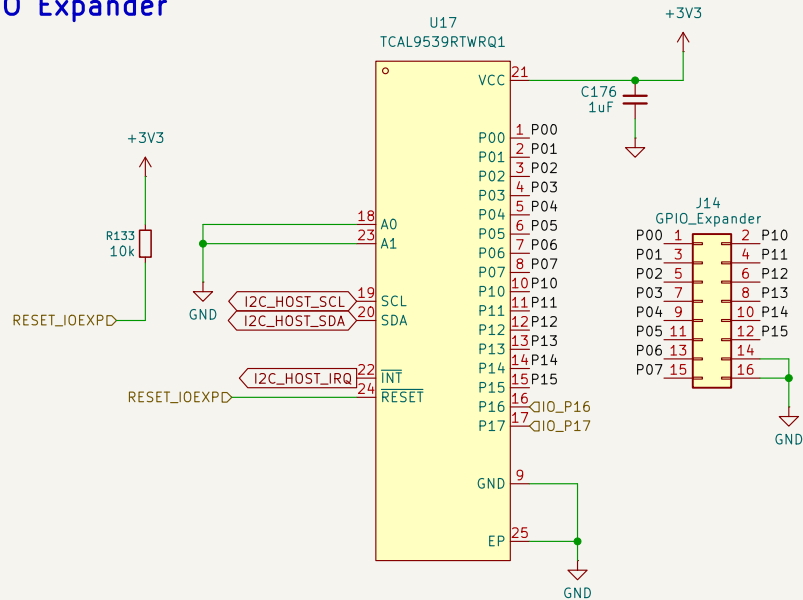
Rev:
 Id: 7/8

Individually Addressable RGB LEDs

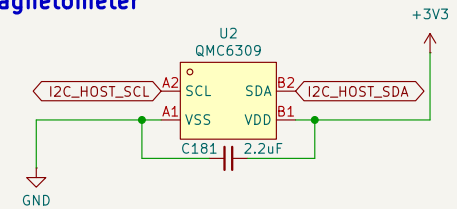
For use with WS2813/WS2816



GPIO Expander



Magnetometer



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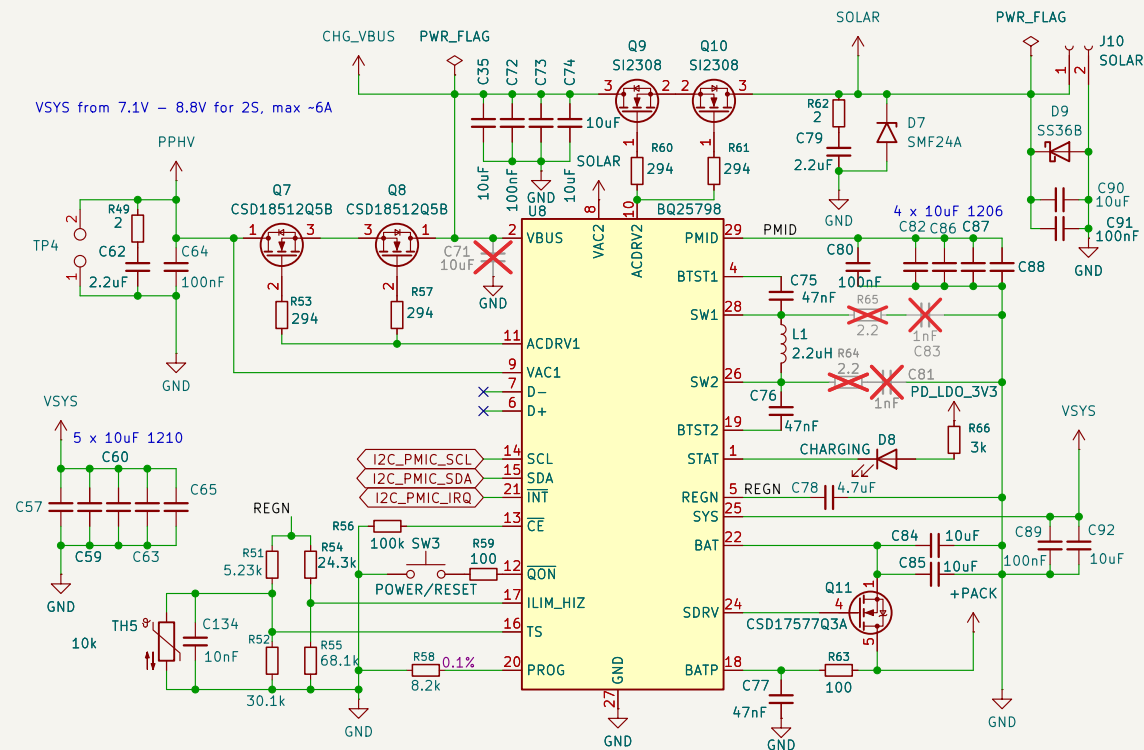
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File: peripherals.kicad_sch

Title: GPIO Expander / LEDs / Magnetometer

Size: A4 Date: 2026-09-24
KiCad E.D.A. 10.99.0-4368-g0839c03fd1

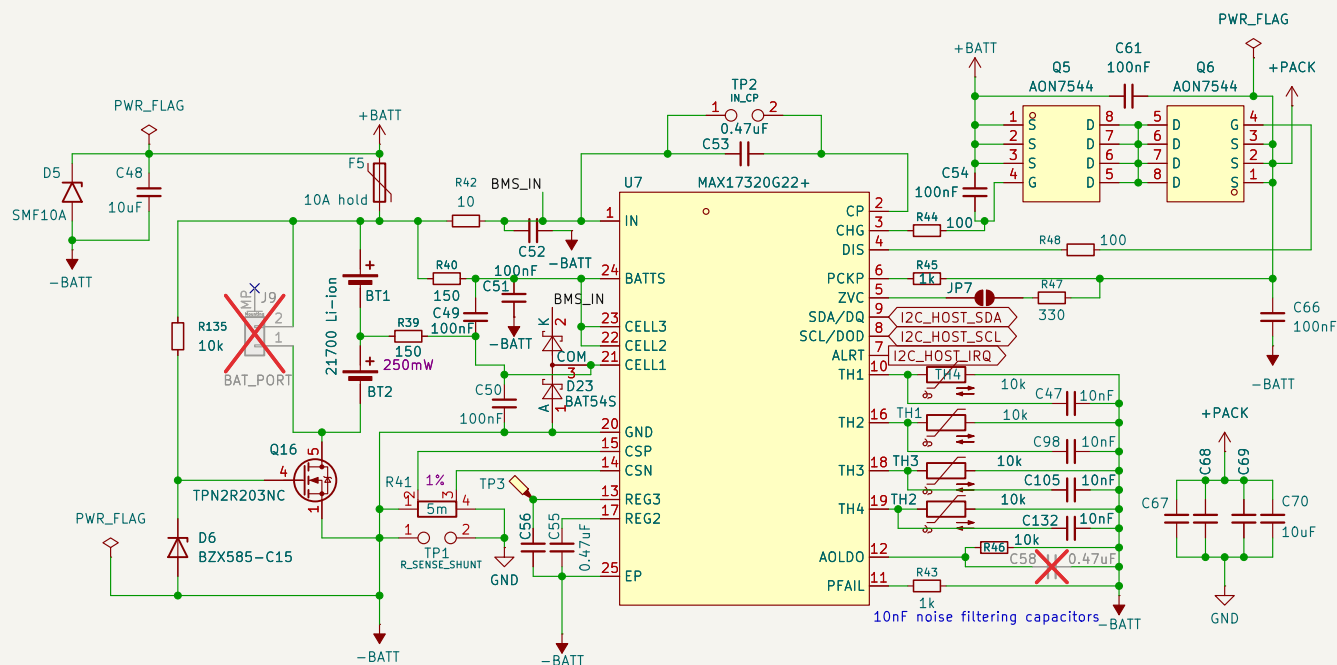
Rev:
Id: 8/8

Capacitors voltage ratings specified,
generally >25V – use XR7/C0G



Id: 3/8

Battery Management System (BMS)



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Sheet: /BMS/

File: bms.kicad_sch

Title: Battery Managment System (BMS)

Size: A4 Date: 2026-09-22

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Rev:

Id: 4/8